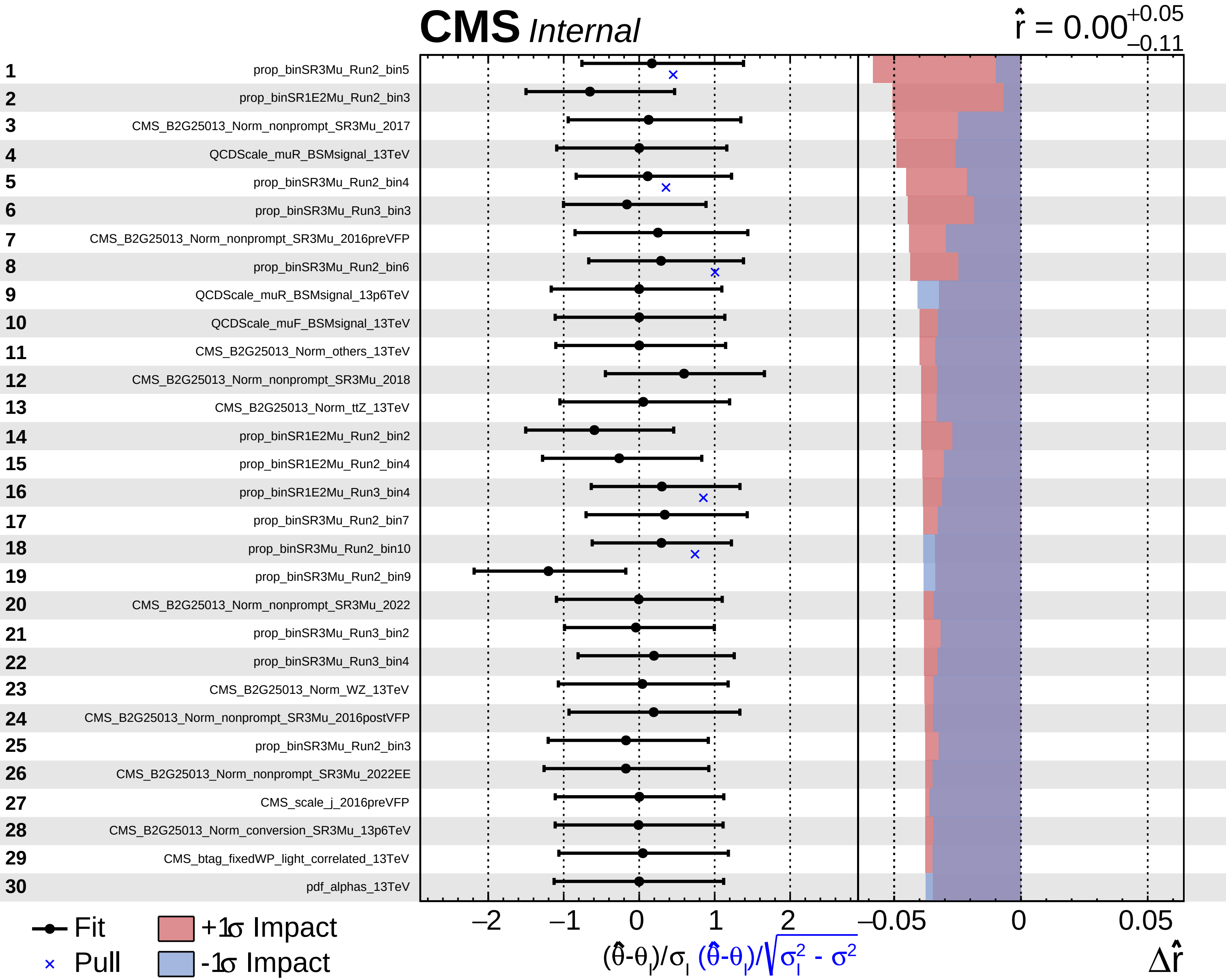
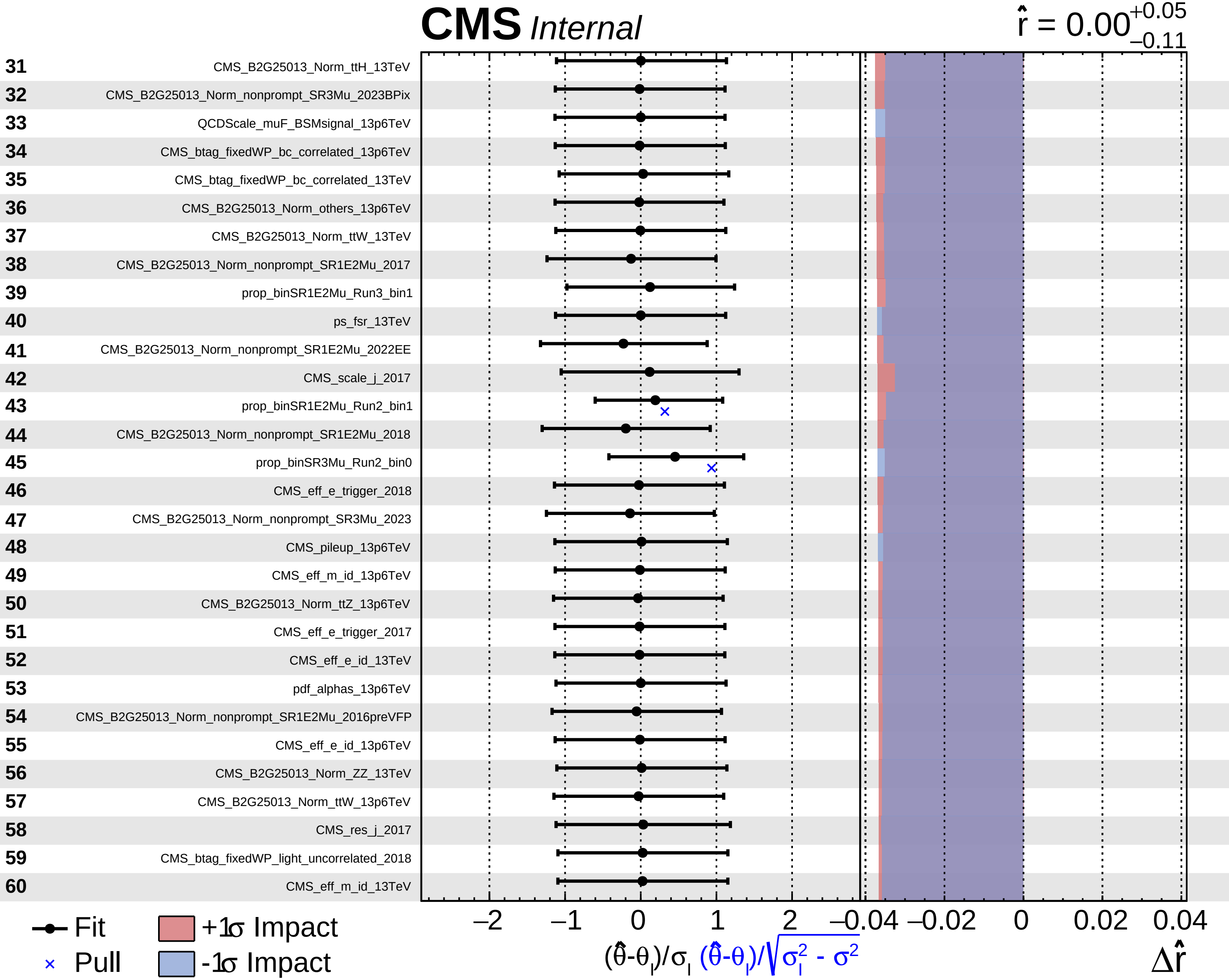
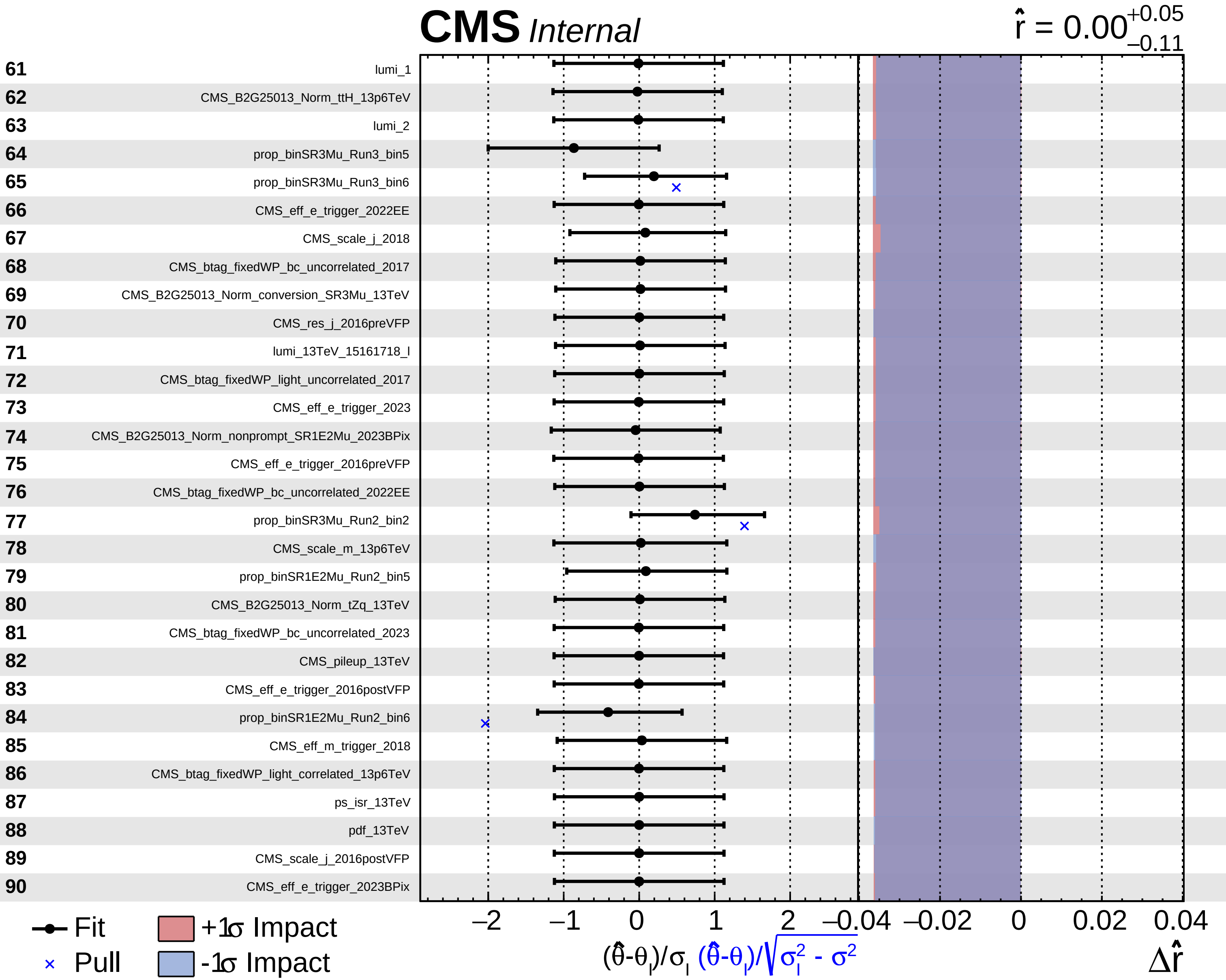
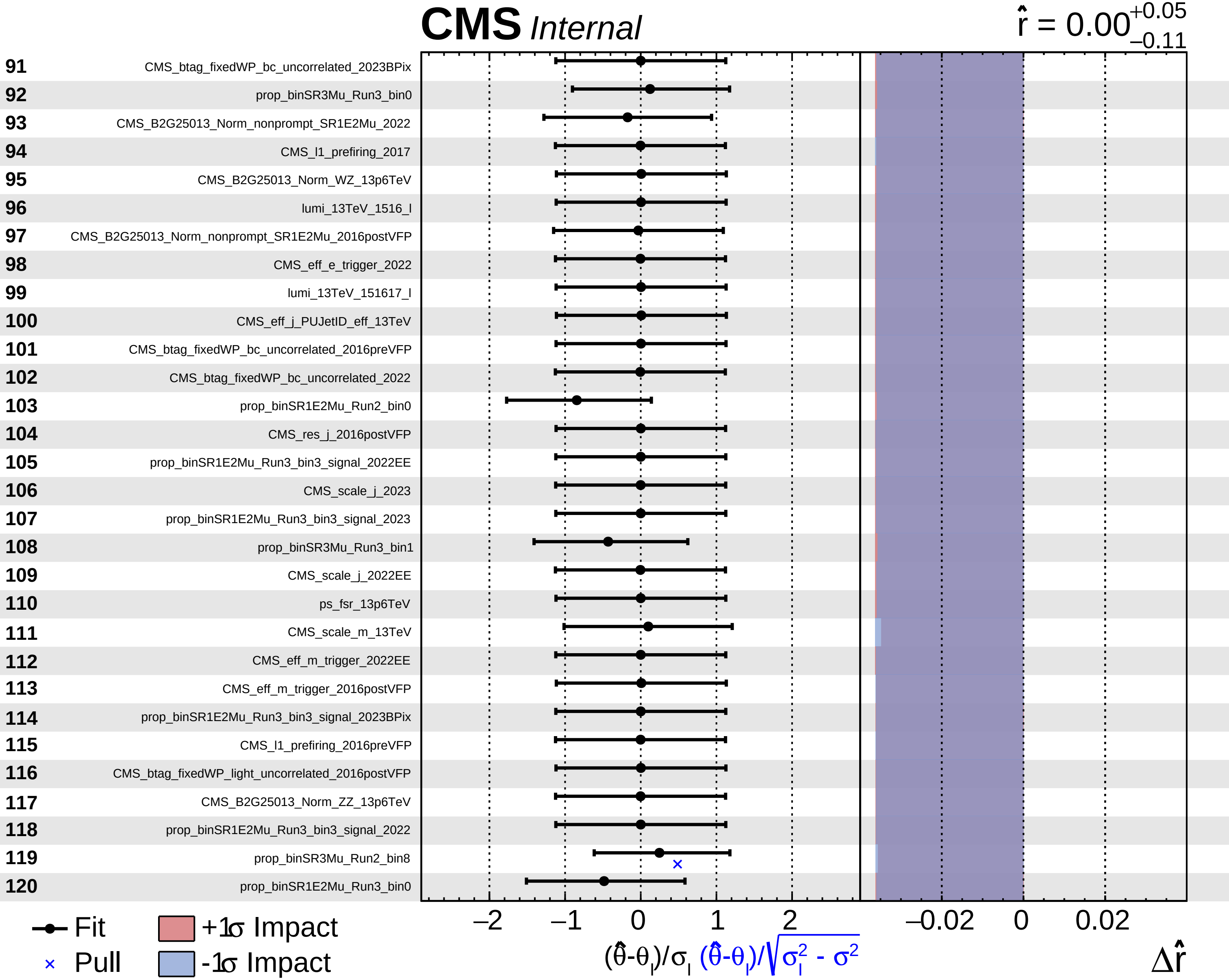


CMS Internal
$\hat{r} = 0.00^{+0.05}_{-0.11}$
1 prop_binSR3Mu_Run2_bin5
2 prop_binSR1E2Mu_Run2_bin3
3 CMS_B2G25013_Norm_nonprompt_SR3Mu_2017
4 QCDScale_muR_BSMsignal_13TeV
5 prop_binSR3Mu_Run2_bin4
6 prop_binSR3Mu_Run3_bin3
7 CMS_B2G25013_Norm_nonprompt_SR3Mu_2016preVFP
8 prop_binSR3Mu_Run2_bin6
9 QCDScale_muR_BSMsignal_13p6TeV
10 QCDScale_muF_BSMsignal_13TeV
11 CMS_B2G25013_Norm_others_13TeV
12 CMS_B2G25013_Norm_nonprompt_SR3Mu_2018
13 CMS_B2G25013_Norm_ttZ_13TeV
14 prop_binSR1E2Mu_Run2_bin2
15 prop_binSR1E2Mu_Run2_bin4
16 prop_binSR1E2Mu_Run3_bin4
17 prop_binSR3Mu_Run2_bin7
18 prop_binSR3Mu_Run2_bin10
19 prop_binSR3Mu_Run2_bin9
20 CMS_B2G25013_Norm_nonprompt_SR3Mu_2022
21 prop_binSR3Mu_Run3_bin2
22 prop_binSR3Mu_Run3_bin4
23 CMS_B2G25013_Norm_WZ_13TeV
24 CMS_B2G25013_Norm_nonprompt_SR3Mu_2016postVFP
25 prop_binSR3Mu_Run2_bin3
26 CMS_B2G25013_Norm_nonprompt_SR3Mu_2022EE
27 CMS_scale_j_2016preVFP
28 CMS_B2G25013_Norm_conversion_SR3Mu_13p6TeV
29 CMS_btag_fixedWP_light_correlated_13TeV
30 pdf_alphas_13TeV
−2 −1 0 1 2
−0.05 0 0.05
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

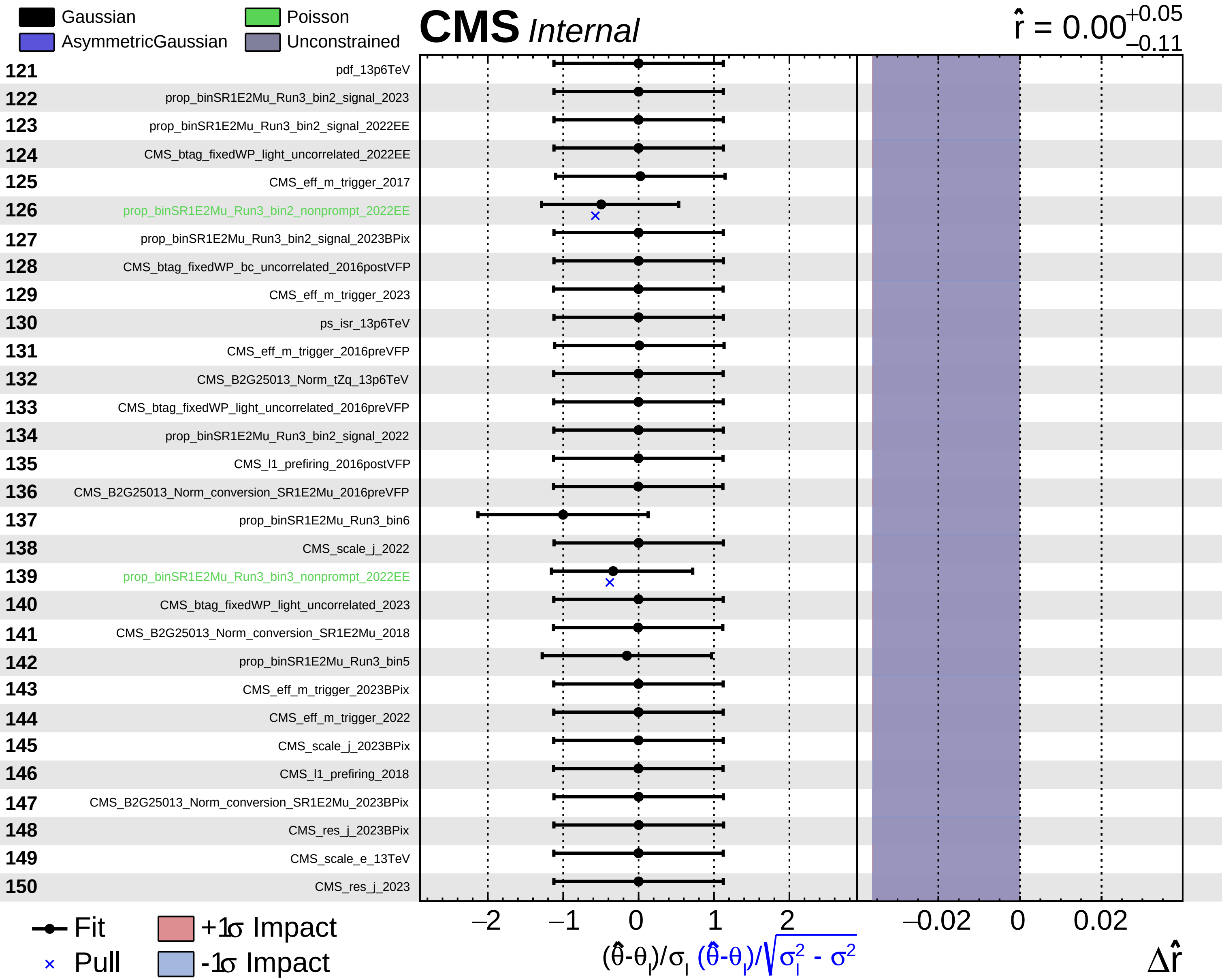


CMS Internal
r̂ = 0.00 +0.05 −0.11
31 CMS_B2G25013_Norm_ttH_13TeV
32 CMS_B2G25013_Norm_nonprompt_SR3Mu_2023BPix
33 QCDScale_muF_BSMsignal_13p6TeV
34 CMS_btag_fixedWP_bc_correlated_13p6TeV
35 CMS_btag_fixedWP_bc_correlated_13TeV
36 CMS_B2G25013_Norm_others_13p6TeV
37 CMS_B2G25013_Norm_ttW_13TeV
38 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2017
39 prop_binSR1E2Mu_Run3_bin1
40 ps_fsr_13TeV
41 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2022EE
42 CMS_scale_j_2017
43 prop_binSR1E2Mu_Run2_bin1
44 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2018
45 prop_binSR3Mu_Run2_bin0
46 CMS_eff_e_trigger_2018
47 CMS_B2G25013_Norm_nonprompt_SR3Mu_2023
48 CMS_pileup_13p6TeV
49 CMS_eff_m_id_13p6TeV
50 CMS_B2G25013_Norm_ttZ_13p6TeV
51 CMS_eff_e_trigger_2017
52 CMS_eff_e_id_13TeV
53 pdf_alphas_13p6TeV
54 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2016preVFP
55 CMS_eff_e_id_13p6TeV
56 CMS_B2G25013_Norm_ZZ_13TeV
57 CMS_B2G25013_Norm_ttW_13p6TeV
58 CMS_res_j_2017
59 CMS_btag_fixedWP_light_uncorrelated_2018
60 CMS_eff_m_id_13TeV
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

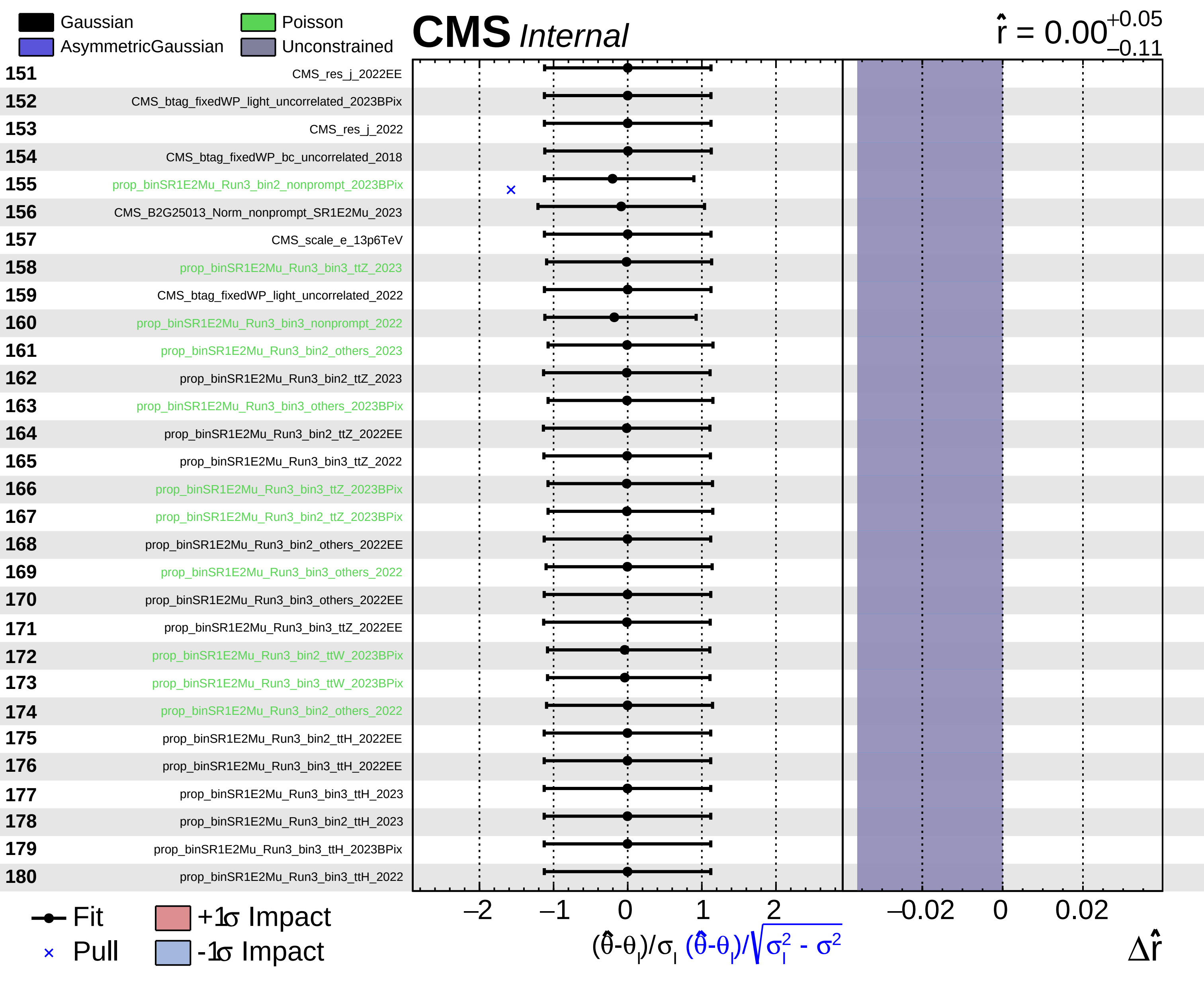

CMS Internal
r̂ = 0.00 +0.05 −0.11
61 lumi_1
62 CMS_B2G25013_Norm_ttH_13p6TeV
63 lumi_2
64 prop_binSR3Mu_Run3_bin5
65 prop_binSR3Mu_Run3_bin6
66 CMS_eff_e_trigger_2022EE
67 CMS_scale_j_2018
68 CMS_btag_fixedWP_bc_uncorrelated_2017
69 CMS_B2G25013_Norm_conversion_SR3Mu_13TeV
70 CMS_res_j_2016preVFP
71 lumi_13TeV_15161718_l
72 CMS_btag_fixedWP_light_uncorrelated_2017
73 CMS_eff_e_trigger_2023
74 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2023BPix
75 CMS_eff_e_trigger_2016preVFP
76 CMS_btag_fixedWP_bc_uncorrelated_2022EE
77 prop_binSR3Mu_Run2_bin2
78 CMS_scale_m_13p6TeV
79 prop_binSR1E2Mu_Run2_bin5
80 CMS_B2G25013_Norm_tZq_13TeV
81 CMS_btag_fixedWP_bc_uncorrelated_2023
82 CMS_pileup_13TeV
83 CMS_eff_e_trigger_2016postVFP
84 prop_binSR1E2Mu_Run2_bin6
85 CMS_eff_m_trigger_2018
86 CMS_btag_fixedWP_light_correlated_13p6TeV
87 ps_isr_13TeV
88 pdf_13TeV
89 CMS_scale_j_2016postVFP
90 CMS_eff_e_trigger_2023BPix
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

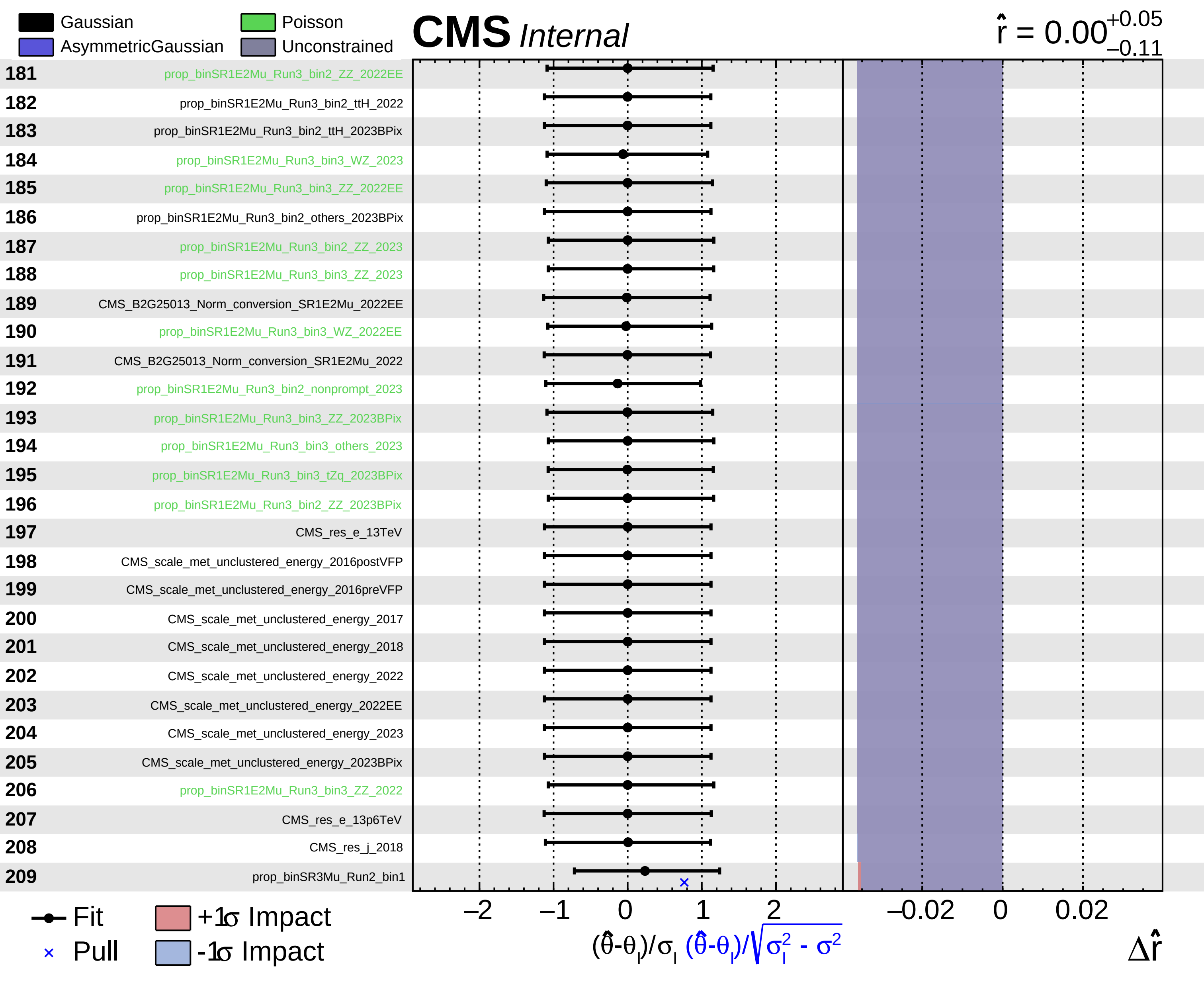


CMS Internal
$\hat{r} = 0.00^{+0.05}_{-0.11}$
91 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
92 prop_binSR3Mu_Run3_bin0
93 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2022
94 CMS_l1_prefiring_2017
95 CMS_B2G25013_Norm_WZ_13p6TeV
96 lumi_13TeV_1516_l
97 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2016postVFP
98 CMS_eff_e_trigger_2022
99 lumi_13TeV_151617_l
100 CMS_eff_j_PUJetID_eff_13TeV
101 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
102 CMS_btag_fixedWP_bc_uncorrelated_2022
103 prop_binSR1E2Mu_Run2_bin0
104 CMS_res_j_2016postVFP
105 prop_binSR1E2Mu_Run3_bin3_signal_2022EE
106 CMS_scale_j_2023
107 prop_binSR1E2Mu_Run3_bin3_signal_2023
108 prop_binSR3Mu_Run3_bin1
109 CMS_scale_j_2022EE
110 ps_fsr_13p6TeV
111 CMS_scale_m_13TeV
112 CMS_eff_m_trigger_2022EE
113 CMS_eff_m_trigger_2016postVFP
114 prop_binSR1E2Mu_Run3_bin3_signal_2023BPix
115 CMS_l1_prefiring_2016preVFP
116 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
117 CMS_B2G25013_Norm_ZZ_13p6TeV
118 prop_binSR1E2Mu_Run3_bin3_signal_2022
119 prop_binSR3Mu_Run2_bin8
120 prop_binSR1E2Mu_Run3_bin0
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 0.00^{+0.05}_{-0.11}$
121 pdf_13p6TeV
122 prop_binSR1E2Mu_Run3_bin2_signal_2023
123 prop_binSR1E2Mu_Run3_bin2_signal_2022EE
124 CMS_btag_fixedWP_light_uncorrelated_2022EE
125 CMS_eff_m_trigger_2017
126 prop_binSR1E2Mu_Run3_bin2_nonprompt_2022EE
127 prop_binSR1E2Mu_Run3_bin2_signal_2023BPix
128 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
129 CMS_eff_m_trigger_2023
130 ps_isr_13p6TeV
131 CMS_eff_m_trigger_2016preVFP
132 CMS_B2G25013_Norm_tZq_13p6TeV
133 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
134 prop_binSR1E2Mu_Run3_bin2_signal_2022
135 CMS_l1_prefiring_2016postVFP
136 CMS_B2G25013_Norm_conversion_SR1E2Mu_2016preVFP
137 prop_binSR1E2Mu_Run3_bin6
138 CMS_scale_j_2022
139 prop_binSR1E2Mu_Run3_bin3_nonprompt_2022EE
140 CMS_btag_fixedWP_light_uncorrelated_2023
141 CMS_B2G25013_Norm_conversion_SR1E2Mu_2018
142 prop_binSR1E2Mu_Run3_bin5
143 CMS_eff_m_trigger_2023BPix
144 CMS_eff_m_trigger_2022
145 CMS_scale_j_2023BPix
146 CMS_l1_prefiring_2018
147 CMS_B2G25013_Norm_conversion_SR1E2Mu_2023BPix
148 CMS_res_j_2023BPix
149 CMS_scale_e_13TeV
150 CMS_res_j_2023
−2 −1 0 1 2
−0.02 0 0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 0.00^{+0.05}_{-0.11}$
151 CMS_res_j_2022EE
152 CMS_btag_fixedWP_light_uncorrelated_2023BPix
153 CMS_res_j_2022
154 CMS_btag_fixedWP_bc_uncorrelated_2018
155 prop_binSR1E2Mu_Run3_bin2_nonprompt_2023BPix
156 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2023
157 CMS_scale_e_13p6TeV
158 prop_binSR1E2Mu_Run3_bin3_ttZ_2023
159 CMS_btag_fixedWP_light_uncorrelated_2022
160 prop_binSR1E2Mu_Run3_bin3_nonprompt_2022
161 prop_binSR1E2Mu_Run3_bin2_others_2023
162 prop_binSR1E2Mu_Run3_bin2_ttZ_2023
163 prop_binSR1E2Mu_Run3_bin3_others_2023BPix
164 prop_binSR1E2Mu_Run3_bin2_ttZ_2022EE
165 prop_binSR1E2Mu_Run3_bin3_ttZ_2022
166 prop_binSR1E2Mu_Run3_bin3_ttZ_2023BPix
167 prop_binSR1E2Mu_Run3_bin2_ttZ_2023BPix
168 prop_binSR1E2Mu_Run3_bin2_others_2022EE
169 prop_binSR1E2Mu_Run3_bin3_others_2022
170 prop_binSR1E2Mu_Run3_bin3_others_2022EE
171 prop_binSR1E2Mu_Run3_bin3_ttZ_2022EE
172 prop_binSR1E2Mu_Run3_bin2_ttW_2023BPix
173 prop_binSR1E2Mu_Run3_bin3_ttW_2023BPix
174 prop_binSR1E2Mu_Run3_bin2_others_2022
175 prop_binSR1E2Mu_Run3_bin2_ttH_2022EE
176 prop_binSR1E2Mu_Run3_bin3_ttH_2022EE
177 prop_binSR1E2Mu_Run3_bin3_ttH_2023
178 prop_binSR1E2Mu_Run3_bin2_ttH_2023
179 prop_binSR1E2Mu_Run3_bin3_ttH_2023BPix
180 prop_binSR1E2Mu_Run3_bin3_ttH_2022
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 0.00 +0.05 −0.11
181 prop_binSR1E2Mu_Run3_bin2_ZZ_2022EE
182 prop_binSR1E2Mu_Run3_bin2_ttH_2022
183 prop_binSR1E2Mu_Run3_bin2_ttH_2023BPix
184 prop_binSR1E2Mu_Run3_bin3_WZ_2023
185 prop_binSR1E2Mu_Run3_bin3_ZZ_2022EE
186 prop_binSR1E2Mu_Run3_bin2_others_2023BPix
187 prop_binSR1E2Mu_Run3_bin2_ZZ_2023
188 prop_binSR1E2Mu_Run3_bin3_ZZ_2023
189 CMS_B2G25013_Norm_conversion_SR1E2Mu_2022EE
190 prop_binSR1E2Mu_Run3_bin3_WZ_2022EE
191 CMS_B2G25013_Norm_conversion_SR1E2Mu_2022
192 prop_binSR1E2Mu_Run3_bin2_nonprompt_2023
193 prop_binSR1E2Mu_Run3_bin3_ZZ_2023BPix
194 prop_binSR1E2Mu_Run3_bin3_others_2023
195 prop_binSR1E2Mu_Run3_bin3_tZq_2023BPix
196 prop_binSR1E2Mu_Run3_bin2_ZZ_2023BPix
197 CMS_res_e_13TeV
198 CMS_scale_met_unclustered_energy_2016postVFP
199 CMS_scale_met_unclustered_energy_2016preVFP
200 CMS_scale_met_unclustered_energy_2017
201 CMS_scale_met_unclustered_energy_2018
202 CMS_scale_met_unclustered_energy_2022
203 CMS_scale_met_unclustered_energy_2022EE
204 CMS_scale_met_unclustered_energy_2023
205 CMS_scale_met_unclustered_energy_2023BPix
206 prop_binSR1E2Mu_Run3_bin3_ZZ_2022
207 CMS_res_e_13p6TeV
208 CMS_res_j_2018
209 prop_binSR3Mu_Run2_bin1
−2 −1 0 1 2
−0.02 0 0.02
Fit
+1σ Impact
Pull
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$